

Data Science Quiz

Pandas • Matplotlib • Seaborn | Time: 60 minutes | Total: 60 marks

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Instructions

- Open the **Quiz_Starter.ipynb** notebook in Google Colab and run the first cell. It loads the dataset as `df`.
- You **may** use the cheat sheet. This quiz tests your thinking, not your memory.
- Write your code in a new cell under each question number. Run it and make sure it works!
- For plots, always add a **title** and **axis labels** - marks are given for these.
- If you get stuck, write down what you *tried* - partial marks are given for good ideas.
- Suggested time: Section A - 8 min, Section B - 22 min, Section C - 14 min, Section D - 14 min, Bonus - if time is left.

The Dataset: video_games.csv

40 video games with 9 columns. Here is what each column means:

Column	What it means	Example
Title	Name of the game	Kart Kings
Genre	Type of game	Racing, RPG, Puzzle...
Platform	Where you play it	Switch, PC, Xbox, PlayStation, Mobile
Release_Year	Year it came out	2019
Rating	Review score out of 10 (has some missing values!)	8.7
Price	Price in dollars (0.00 = free game)	39.99
Copies_Sold_Millions	How many million copies were sold	21.5
Hours_To_Beat	Hours to finish the game (has a missing value!)	12
Multiplayer	Can you play with friends?	Yes / No

Tip: Before starting, run `df.head()`, `df.info()` and `df.describe()` to get to know the data.

Section A - Multiple Choice (6 x 1 = 6 marks)

Circle or write the letter of the correct answer. You may test your idea in Colab first!

A1. What does `df.shape` return for our dataset?

- (a) (9, 40)
- (b) (40, 9)
- (c) 40
- (d) 360

A2. Which line gives the **average** Rating of all games?

- (a) `df['Rating'].sum()`
- (b) `df['Rating'].average()`
- (c) `df['Rating'].mean()`
- (d) `df.mean('Rating')`

A3. `df[df['Price'] == 0]` will give you...

- (a) All games with a missing price
- (b) All free games
- (c) The first row only
- (d) An error

A4. You want to see how many games are on each Platform. Which is the BEST tool?

- (a) `df['Platform'].value_counts()`
- (b) `df['Platform'].sum()`
- (c) `df.describe()`
- (d) `df['Platform'].mean()`

A5. Which plot is best for showing the **relationship between two number columns** (like Price and Rating)?

- (a) Bar chart
- (b) Pie chart
- (c) Scatter plot
- (d) Histogram

A6. What does `df.dropna()` do?

- (a) Deletes all columns
- (b) Removes rows that have missing values
- (c) Fills missing values with 0
- (d) Sorts the data

Section B - Pandas Coding (5 questions, 22 marks)

Write code for each. Print or display your answer so it is visible.

B1. Explore the data [3 marks]

Write code to find out: (a) how many rows and columns the dataset has, (b) which columns have missing values and how many, (c) the name of the game with the **highest** Rating.

B2. Filter like a detective [4 marks]

Show only the games that are **Multiplayer**, cost **less than \$40**, and have a Rating of **8 or higher**. How many games match? Display just their Title, Price and Rating columns.

B3. Group and compare [5 marks]

(a) Find the **average Copies_Sold_Millions** for each **Genre**, sorted from highest to lowest. (b) Which Platform has the **most games** in the dataset? (c) For each Platform, what is the **most expensive** game price?

B4. Create new columns [5 marks]

(a) Add a new column `Revenue_Millions = Price x Copies_Sold_Millions`. (b) Add a column `Age` = how many years old the game is (the current year is 2026). (c) Add a column `Is_Free` that is True when Price is 0, otherwise False. (d) Show the top 5 games by `Revenue_Millions`.

B5. Clean the data [5 marks]

The `Rating` column has missing values, and the game 'Block Builder' has a strange `Hours_To_Beat` of 999. (a) Fill the missing Ratings with the **average** Rating. (b) Fill the missing `Hours_To_Beat` with the **median**. (c) Remove the row where `Hours_To_Beat` is 999 (it is an outlier). (d) Explain in one sentence: *why did you use the median instead of the mean for Hours_To_Beat?*

Section C - Matplotlib (3 questions, 14 marks)

Every plot needs a title and axis labels. Use `plt.show()` at the end.

C1. Bar chart [4 marks]

Make a **bar chart** showing the **number of games per Genre**. Rotate the x-axis labels so they don't overlap. Give it a fun title.

C2. Scatter plot [5 marks]

Make a **scatter plot** of Price (x-axis) vs Copies_Sold_Millions (y-axis). Then answer in a comment: *do cheaper games sell more copies? What do you notice about the free games?*

C3. Two plots side by side [5 marks]

Using `plt.subplots(1, 2, figsize=(12, 4))`, make two plots next to each other: (left) a **histogram** of Rating with 10 bins, (right) a **line plot** of how many games were released each year (hint: `value_counts().sort_index()`). Give each subplot its own title.

Section D - Seaborn (3 questions, 14 marks)

D1. Count plot [4 marks]

Make a seaborn **countplot** of Platform, using `hue='Multiplayer'`. In a comment, say which platform has the most single-player games.

D2. Box plot [5 marks]

Make a **boxplot** of Rating for each Genre. Then answer in a comment: *which genre has the highest median rating? Which genre has the widest spread?*

D3. Heatmap [5 marks]

(a) Make a correlation table of the number columns using `df.corr(numeric_only=True)`. (b) Draw it as a seaborn **heatmap** with `annot=True`. (c) In a comment, name the two columns that are **most strongly correlated** and say whether it is positive or negative.

Bonus - Be a Data Scientist (4 marks, optional)

A game studio asks you: **"We want to make a new game that sells a LOT of copies. What should we make?"**

Use the data (any pandas/plots you like) to give them advice. Write 2-3 sentences with your recommendation for **Genre**, **Platform**, **Price** and **Multiplayer or not**, and say which numbers convinced you.

Good luck! Remember: a plot that is clear and labelled beats a fancy one nobody understands.